CHARAN TEJA VALAVALA

B.E. COMPUTER SCIENCE ENGINEERING | APPLIED AI

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GitHub | Portfolio | LinkedIn

PROFILE

Computer Science Engineering undergraduate building evidence-grounded retrieval and applied AI systems, with web-development experience supporting a learning platform used by 200+ students and trainees.

PROJECTS

Course Copilot *Python | FastAPI | MiniLM | SQLite FTS5*

- Built hybrid document retrieval with MiniLM embeddings, FTS5 BM25, reciprocal-rank fusion, PDF/TXT ingestion, and source/page evidence; optional Ollama answers require valid evidence citation numbers.
- Ablated BM25-only, dense-only, and fused retrieval on 67 hand-authored queries; 2:1 BM25:dense fusion reached Hit@1 0.923, Hit@5 1.000, and MRR 0.962 on 52 answerable queries, with 14/15 correct out-of-scope abstentions.

FormScope *Python | FastAPI | SEC EDGAR | XBRL*

- Built a source-linked filings explorer that validates annual 10-K periods, merges fallback XBRL concepts, computes revenue growth and net margin, and ranks primary-filing passages.
- Cross-checked annual revenue and net income against inline XBRL from 20 issuers; all 40 filing facts matched FormScope's SEC Company Facts output.

IncidentPilot *Python | FastAPI | SQLite | Ollama*

- Built a stateful incident-triage simulator with read-only log, metric, and runbook tools, evidence-linked diagnoses, versioned event traces, and approval-gated simulated actions.
- Added a 20-case synthetic Ollama evaluation harness for tool-call validity, evidence-reference validity, model-only diagnosis accuracy, and policy fallback; no model score is claimed.

ACADEMIC PROJECTS

Diabetic Retinopathy Detection *Python | OpenCV | scikit-learn*

- Built retinal fundus-image classifiers using texture and color features with SVMs; evaluated accuracy, precision, recall, and sensitivity.

DRL for Object Tracking *Monte Carlo control | stochastic grid environment*

- Implemented a grid environment with stochastic thermal states and teleportation; compared on-policy and off-policy Monte Carlo control and analyzed convergence and policy optimization.

EXPERIENCE

MDDTI SBC | Web Development Intern May-July 2024

- Built HTML/CSS pages for the organization website and contributed to a MongoDB-backed online test platform used by 200+ students and trainees.

EDUCATION

BITS Pilani, Hyderabad Campus | B.E. Computer Science Engineering | CGPA 5.2/10 Expected 2027+

AP EAMCET | Statewide rank 8

Narayana Junior College | Class XII | 98.1% 2022

Narayana High School | Class X | 10/10 GPA 2020

TECHNICAL SKILLS

Programming: Python, JavaScript, HTML, CSS
Coursework languages: C, C++, Java
Frameworks and data: FastAPI, SQLite, MongoDB, Sentence Transformers, FTS5/BM25, SEC EDGAR/XBRL
Tools: Git, Docker, Ollama, OpenCV, scikit-learn

RELEVANT COURSEWORK

Artificial Intelligence, Machine Learning, Reinforcement Learning, Foundations of Data Science, Data Structures and Algorithms, Database Systems, Object-Oriented Programming

LANGUAGES

English, Hindi, Telugu